COLOSSEUM · SOLANA

Vitals

A patient is dying on a clock and you decide what happens next.
Anyone can play. Nobody can fake the replay.

VERIFIABLE CLINICAL REPLAY RUST · SOLANA AGPL-3.0 + COMMERCIAL

THE PROBLEM

Skill has no passport.

Since 2024, ECFMG certification — the gate to US residency — requires graduating from a school accredited by a WFME-recognised agency. Accreditation is granted to *institutions*, country by country, and most countries do not have it.

4,350+	~23	130–147
medical schools in the World Directory	countries with a WFME-recognised agency	countries whose graduates apply

Every one of those graduates ran hundreds of scored clinical encounters that would have proved what they can do. All of it sits in one institution's database, unverifiable by anyone outside it, and is thrown away at the border. **The evidence exists. The trust layer does not.**

NOT A MOCKUP

This runs. Here is the output.

One scenario — anaphylaxis, a 19-year-old, ten minutes from the first bite. Three players, three tapes, replayed against the shipped physiology automaton.

treated	adrenaline IM · O_2 · supine · fluids · admit	**WinDischarge**
stood up	adrenaline IM · **stood the patient up** · O_2 · fluids · admit	**WinDischarge · harm**
no adrenaline	antihistamine · steroid · wait	**DeathArrest**

The middle row is the one that matters. **Same outcome as the first, different leaf**, because the harm is on the record. The anchor captures how you got there, not just whether you won. And nobody scripted the third — antihistamines are what people reach for when they haven't recognised what they're looking at, and the simulation kills her for it.

```
sce_hash  4d5ee49cc13dd5af1badf4d601010b03e70ded6d2e0ebaf339a26e7d1616a703
treated   65c56e52d9f718aef60cb4e7f712443fe779b10ffb3c3ac2e301464ebd275398
stood up  a6bec396869e7d889686bc42111f4a2b090aa16c4d93213262d24c858637f8ca
```

THE MECHANISM

Every run reduces to one hash.

1 · COMMIT

Before play: `hash(scenario ‖ player ‖ nonce)` onchain. The chain knows how many attempts were *started*, so grinding for a good run is visible.

2 · PLAY

A deterministic automaton — vital dynamics, triggers, transitions, harm, terminal outcome. The tape is what you did and when.

3 · REDUCE

Discrete facts only: ordered beats, harm events, outcome. No float trajectory enters the hash, so nothing drifts across machines.

4 · ANCHOR

One compressed leaf. Anyone holding the tape and the scenario recomputes it — forever, without us.

```
leaf = sha256( sce_hash ‖ tape ‖ beats ‖ harm_events ‖ outcome )
```

There is no model anywhere in that path. An LLM plays the patient's dialogue in other modes and never touches the evidence. This is verifiable replay — what chess engines and speedrun leaderboards have done for decades — applied to a patient who is dying on a clock.

WHY THIS NEEDS A CHAIN

Watch the program refuse its own user.

Progression is not granted by a server. The program is handed the attempts, **recomputes the level itself**, and writes only what its own arithmetic agrees with. Live on a validator, today:

CLAIM: PROFICIENT

```
REJECTED — claim rejected: claimed Proficient, computed Competent
from 3 attempts (3 distinct, avg 7333bps)
```

CLAIM: COMPETENT

GRANTED — stored level Competent, distinct 3, xp 35. The level on chain is the one the program computed, never the one that was claimed.

No issuer key. No oracle. No authority that could be leaned on, bought, or subpoenaed. Take the chain away and you are back to trusting one company's database — which is precisely why digital credentials are worthless today.

ONCHAIN ARCHITECTURE

Four primitives, none of them invented here.

LAYER	WHAT IT DOES	SOLANA PRIMITIVE
Scenario registry	Authors publish; content stays off-chain, only hash + price + royalty split onchain	Anchor program · Token-2022
Replay anchor	One compressed leaf per run — the high-volume write	Bubblegum state compression
Progression	Levels, skill trees, badges — minted permissionlessly, predicate recomputed onchain	Token-2022 *NonTransferable*
Credential	Accredited issuer attests an EPA competency to the player's wallet	Attestation Service

VOLUME

A single cohort year across the World Directory is hundreds of millions of runs. Compression puts that at **~$110 per million**. Anchoring only the exams would be the cheap-chain compromise and would kill the thesis.

LATENCY

The skill tree advances in the same screen as the outcome. On a slow chain that becomes a pending spinner and stops feeling like a result.

ONBOARDING

Fee-payer relay. A player never sees a seed phrase, never holds SOL, never learns any of this is happening.

TOKENOMICS

The token secures verification. It never represents competence.

Someone has to replay the tape and attest the outcome. Make that one party and you have rebuilt the database we are trying to escape. **$VITAL is what makes verification permissionless and expensive to lie in.**

THE LOOP

STAKE

Operators bond $VITAL to accept replay work. Bond size sets how much throughput they can take.

VERIFY

Replay is deterministic, so **two honest verifiers must agree**. Fees are paid by institutions and sponsors — in USDC, never by players.

DISPUTE → SLASH

Disagreement is provable: anyone posts a bond and re-runs the tape. The wrong verifier is slashed, the challenger is paid from it. Fraud proofs work here because the computation is exact.

SINKS · SOURCES

sink	verifier bond (locked while active)	**supply out**
sink	scenario listing + dispute bonds	**supply out**
sink	slashing — burn share	**burned**
source	verification fees → operators	**earned**
source	replay royalties → scenario authors	**earned**
note	authoring is cheap, so the author market is real	—

NEVER TOKENIZED

- **Credentials, badges, skill trees.** Token-2022 NonTransferable, no exceptions — a tradeable "Expert in Cardiology" is credential fraud with extra steps.
- **Access to play.** No token gates a student out of practice. Ever.

Allocation is a proposal, not a commitment, and no token ships inside the hackathon window: the verification market only has to exist once verification is worth attacking. Distribution weighted to operators and scenario authors — the two parties whose work the network actually consumes.

MARKET AND PRECEDENT

This shape already won this hackathon once.

FRONTIER 2026 · GRAND CHAMPION

CrowdBrain — "a vertically-integrated robotics DePIN built to **train users in simulation, qualify them through QA, and route the best operators to real robots**."

Train in simulation → verify competence → route the qualified to real work. That is Vitals, for robot operators instead of clinicians. We think the pattern generalises; the largest crypto hackathon ever run agreed three months ago, out of 2,857 submissions from 150 countries.

HOW EPISODE SIX GETS MADE

Stills from an image model, then **a critic crew that fails the shot before a video model ever sees it** — a medical advisor checking the procedure, a continuity supervisor checking the face, a director reading both verdicts. Then video. A cinematic episode costs tens of dollars, not a film crew.

It is the same principle as the rest of this: something that must be right, checked automatically. The crew verifies the content. The chain verifies the play.

WHY MEDICINE FIRST

The hardest instance: highest stakes, heaviest regulation, most protective of data. Solve it here and it generalises downward to every other skill credential.

WHO PAYS

Institutions on ordinary invoices, sponsors funding outcome-linked scholarships, and platform fees on the scenario market. Players pay nothing and hold nothing.

HEAD START

A clinical simulator already in production — 424 authored scenarios, deterministic scoring, a hash-chained audit log, real students.

WHERE IT STANDS, AND WHAT'S NEXT

Two halves built. One week to join them.

WORKING NOW

- **Replay → leaf.** EP1 runs against the shipped automaton; three tapes, three distinct leaves, reproducible across runs.
- **Program → refusal.** Native Solana program live on a validator, recomputing levels and rejecting claims that don't survive its own arithmetic.
- **Rust end to end** — one scoring implementation compiled into the game, the verifier, and the program.

THE REMAINING FOUR WEEKS

- **W1** — join the halves: `claim_progress` takes merkle proofs against anchored leaves instead of attempts handed to it.
- **W2** — compressed anchors, scenario registry, public verify page.
- **W3** — soulbound progression, badge-gated scenarios, sponsor escrow paying out on a provable level.
- **W4** — mainnet-beta, demo video, open-source release.

Cut on purpose

ZK selective disclosure · verifier DePIN with live staking · token launch · mobile · new clinical content. Each is a real v2 line and none survives four weeks. The credential layer stays a stub until an accrediting institution is standing next to it — progression doesn't need one, which is why progression is the demo.

The 60-second version: you play a patient who is dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.